1. Definitions & Examples.

① Def: R is a complete intersection of codim. c if $\hat{R} \cong Q/I$, (Q reg. pres.)
I generated by reg. seq. of length c.

Def: R is a hypersurface if it's CI of codim. 1 i.e. $\hat{R} \cong Q/(f)$, $f \in m^2$; conversely $Q/(f)$ is hypersurface.

Note: Any regular local R is CI: $\hat{R}$ is also reg., $\hat{R} = \hat{R}/0$.

② ☆ Method to judge a CI.

$\hat{R} = Q/I \longrightarrow \sqrt{I} = p$ thus $Min(I) = \{p\} \longrightarrow ht(p) = \dim Q - \dim Q/p$
$\longrightarrow$ compare $ht(I)$ & $\mu(I)$.

i) Note that $R = k[[x,y]]/(x^2, xy)$ is already complete: $\widehat{M/IM} = \hat{M}/I\hat{M}$, and power series are complete. (So all quotients of complete rings are complete).

Let $R = k[[x,y]]/(x^2, xy) = Q/I$, this is reg. repn. since $(x^2, xy) \subset m^2$.

- $\sqrt{I} = (x)$ since $(x) \subset \sqrt{I}$, and $t^d \in I \Rightarrow x \mid t^d \Rightarrow x \mid t \Rightarrow \sqrt{I} \subset (x)$.
  Thus $\bigcap_{p \supset I} p = (x)$ prime, and $Min(I) = (x)$.
- By def., $ht(I) = \min\{ht(p) : p \supset I\} = ht(x) = \dim Q - \dim k[[y]] = 1$
  (since Q reg. local $\Rightarrow$ CM)
- But $\mu(I) = 2 \neq 1$ since $x^2, xy \in I$; thus R not CI.

ii) $R = k[[x,y,z]]/(x^2, y^2, z^2, xyz)$ not CI.

Then $\sqrt{I} = m$, $ht(I) = ht(m) = \dim R = 3$, but $\mu(I) = 4 \neq 3$.

2. Koszul homology of R.

① Def: $(R,\mathfrak{m})$ local, $\underline{x}$ any minimal generating set of $\mathfrak{m}$,

$$K^R := Kos^R(\underline{x}).$$

i) $\begin{cases} H_0(K^R) = R/\mathfrak{m} = k \\ H_{\text{embdim}(R)}(K^R) = soc(R) := ann_R(\mathfrak{m}). \end{cases}$

↳ since at $\text{embdim}(R)$, it's R, and $\partial(a e_1 \cdots e_n) = 0 \Leftrightarrow a x_i = 0 \ \forall i$.

ii) Well-definedness: $\underline{x}, \underline{y}$ two minimal gen. sets for $\mathfrak{m}$, then $Kos(\underline{x}) \cong Kos(\underline{y})$.

Pf: Clearly they're isom. dg exterior algebras: min. gen. gives change-of-basis matrix of $\underline{x}, \underline{y}$ is invertible, since as basis in $\mathfrak{m}/\mathfrak{m}^2$, $\det(A) \neq 0 \Rightarrow \notin \mathfrak{m} \Rightarrow$ unit.

iii) ✡ Depth sensitivity:

$$\sup\{i : H_i(K^R) \neq 0\} = \text{embdim}(R) - \text{depth}(R).$$

Note that depth reduces this sup: $\begin{cases} \text{depth}(R) = 0 \Leftrightarrow \mathfrak{m} \in Ass(R) \Rightarrow ann_R(\mathfrak{m}) \neq 0. \\ R \text{ reg.}, \text{ then } H_i(K^R) = 0 \ \forall i > 0. \end{cases}$

iv) Lem: R is regular $\Leftrightarrow H_1(K^R) = 0$.

Pf: R is regular $\Leftrightarrow$ min gen. set of $\mathfrak{m}$, $\underline{x}$ is reg. seq.

$\Leftrightarrow H_1(K^R) = 0$.

② Examples on "the reasons for dgca".

Consider $H(K^{I^2})$ for $R = k[[x,y]]/(x^2, y^2)$ and $k[[x,y]]/(x^2, xy)$.

We compute (x^2, xy).

- Note that $H(K^R)$ is k-alg., while K^R is over R.

$$0 \longrightarrow Re_2 \longrightarrow Re_1 \oplus Re_2 \longrightarrow R \longrightarrow 0,$$

$\partial(e_1) = x$, $\partial(e_2) = y$ since $m = (x,y)$.

- $\partial(ae_1 + be_2) = ax + by \in (x^2, xy) \Rightarrow ax + by = cx^2 + dxy$

$$\Rightarrow x \mid b,\ x \mid a$$

Thus $H_1(K^{I^2})$ is generated by $w_1 = [xe_1]$, $w_2 = [xe_2]$, and B_1 is $xe_2 - ye_1$.

- $mxe_2 - mye_1 = 0 \Rightarrow mx = 0,\ my = 0 \Rightarrow x \mid m$ suffices.

Thus $H_2(K^R) = \langle u \rangle$, $u = [xe_1 e_2]$.

- Therefore: $H_i(K^{I^2}) = \begin{cases} k & i = 0 \\ kw_1 \oplus kw_2 & i = 1 \\ ku & i = 2 \\ 0 & \text{otherw.} \end{cases}$

Note that $R = k[[x,y]]/(x^2, y^2)$ has the same complex structure, their dga structure is not the same: $w_1 w_2 = 0$, while for the other $w_1 w_2 = u$.

(3) ✡✡ Properties of K^R.

i) ✡ $K^{\hat{R}} \simeq K^R$, so works well with completion.

Thm: $\exists$ a quasiisom. of dgas: $K^R \longrightarrow K^{\hat{R}}$.

Pf:

- Two KEY facts: $\hat{-}$ is compatible with quotients & $H(K^R)$ is annihilated

by m.

Then $H(K^R)\otimes_R \hat{R} = H(K^R)\otimes_R \hat{R}/m\hat{R}$

$= H(K^R)\otimes_R k$

$= H(K^R)$ since it's k-v.s., $mH(K^R)=0$

- $\hat{R}$ is flat: thus $\hat{R}$ comutes with homology, and:

$H(K^R) = H(K^R)\otimes_R \hat{R}$

$= H(K_\bullet^R(\underline{x})\otimes_R \hat{R})$

$= H(K_\bullet^{\hat{R}}(\hat{\underline{x}})) = H(K^{\hat{R}})$

where $\hat{\underline{x}}$ is image of x in $\hat{R}$ (f.f.), and clearly $K_\bullet^R(\underline{x})\otimes_R \hat{R}$ is a Koszul complex on $\hat{\underline{x}}$.

ii) Similarly, for any complex of R-mod. M,

if $H_n(M)$ is finite length R-mod, then $M \simeq M\otimes_R \hat{R}$.

pf: this time $H_n/m^{in}H_n = 0$ since H_n fin. length $\Rightarrow$ Artinian, so

$H_n \supsetneq mH_n \supsetneq m^2H_n \supsetneq \cdots$

must terminate. ☆

Then $H_n\otimes_R \hat{R} = H_n\otimes_R \hat{R}/m^{in}\hat{R} = H_n\otimes_{R/m^{in}} R/m^{in} = H_n$

↓ since m-adic completion of R/m^s is 0: $[m]^s = 0$ for $s \gg 0$.

iii) How to compute $H_1(K^R)$.

Note that the result aligns with what we observed: R is reg. $\Leftrightarrow \underline{I} = 0$ in $\hat{R} = Q/\underline{I}$ reg. repn.

a) Thm: Let $R = Q/I$ for $I \subset m^2$, $\underline{y}$ min. gen. for m. Then:

$\exists$ dga isom.: $R \otimes_Q Kos(\underline{y}; Q) \xrightarrow{\simeq} K^R$.

In particular:

$$Tor_i^Q(R, k) = H_i(K^R).$$

Pf:

- Basically one is resolving in Q and one is resolving in quotient $R = Q/I$. Let $\underline{x}$ be image of $\underline{y}$.
- Then $Q^n \otimes_Q R \cong R^n$ induces isom. of complexes

$$Kos^Q(\underline{y}) \otimes_Q R \longrightarrow Kos^R(\underline{x})$$

that's also dga isom.

Taking homology, $Tor_i^Q(R, k) = H_i(K^R)$.

b) Thm: $H_1(K^R) \cong I/mI$, $\mu(H_1(K^R)) = \mu(I)$ (Q/I reg. repn.).

Pf: ✡ May assume $R = \hat{R}$ by ii).

- SES $0 \to I \to Q \to R \to 0$ induces

$$0 \to Tor_1^Q(R, k) \to I \otimes k \xrightarrow{2} Q \otimes k \qquad \text{exact.}$$

Then since $I \subset m$, the inclusion 2 is $0 : I \to 0$ in Q/m.

Thus $Tor_1^Q(R, k) \cong I/mI$.

- By a) $H_1(K^R) = I/mI$.

iv) ✡ Criterion for complete intersection (non two: $Q[x]$ or $H_1(K^R)$).

Thm: for Noeth. local R,

$$\mu(H_1(K^R)) \geq \text{embdim}(R) - \dim(R),$$

and R CI $\Leftrightarrow \mu(H_1(K^R)) = \text{embdim}(R) - \dim(R)$.

Thus the word "codimension".

Pf:

- $ht(I) = \dim Q - \dim(Q/I) = \text{embdim}(R) - \dim(R)$
  - $\dim Q$ since Q CM
  - $\dim(Q/I)$ since $\text{embdim}(Q) = \text{embdim}(R)$
- $\mu(H_1(K^R)) = \mu(I) \geq ht(I) = \text{embdim}(R) - \dim(R)$

& $\mu(I) = ht(I) \Leftrightarrow I$ gen. by reg. seq. $\Leftrightarrow R$ CI.

V) Well-def. of CI.

Coro.: $Q/I \cong \hat{R} \cong S/J$, then:

I is gen. by reg. seq. $\Leftrightarrow J$ is.

Pf:

- May reduce to reg. repn. since if $I = L + (f)$ for $f \in m \setminus m^2$, then:
  - f is reg. on Q; f is min. gen. since $[f] \neq 0$: $f \in mI \subset m^2$ #.
  - Then L is like extending min. gen., and

    I gen. by reg. $\Leftrightarrow L$ gen. by reg. in $Q/(f)$.
- Then
  - $\dim(Q) = \text{embdim}(\hat{R}) = \dim(S) \Rightarrow ht(I) = ht(J)$
  - $\mu(I) = \mu(H_1(K^R)) = \mu(J)$

Hence the result.

3. Another criterion on generation of $H_1(K^R)$.

Thm (Assmus): TAFE:

① R is complete intersection.

② The natural map $\Lambda_k(\Sigma H_1(K^R)) \to H(K^R)$ is isom. of graded alg.

③ $K^R \simeq \Lambda_k(\Sigma H_1(K^R))$ is quasiisom. of local dgcas.

④ $H_2(K^R) = H_1(K^R)^2$ (i.e. H_2 gen. by H_1, and H_1^2 by degree reasons).

$H(M)$ is not naturally a dga.

Pf:

① ⇒ ② ⇒ ③ ⇒ ④.

- Note two isom. ☆☆:

a) $K^R \xleftarrow{\sim} Kos(y,Q)\otimes R \xleftarrow{\sim} Kos(\underline{y},Q)\otimes Q[X] \xrightarrow{\sim} k\otimes Q[X] = k[X]$

by 2.(iii)

res. for $k=Q/m$; res. for $R=Q/I$

{Since tensoring free mod. preserves quasiisom.}

b) $\Lambda_k(\Sigma H_1(K^R)) \cong k[X_1]$.

Note that {both are exterior alg. on $\mu(I)$ variables; ∂_1 are both 0 since $\partial(X_1)=0$ and $\Sigma H_1(K^R)$ has 0 differential.

- Then we have:

$$\begin{array}{ccc} H(K^R) & \xrightarrow{\cong} & H(k[X]) \\ \uparrow & & \uparrow \cong \\ \Lambda_k(\Sigma H_1(K^R)) & \xrightarrow[\cong]{} & k[X_1] \end{array}$$

where $R \subset I$ implies: {$Q[X]$ Koszul, thus $k[X] = k[X_1]$; $\partial_1 = 0$, or $Q[X]$ min. res., so $\partial = 0$, and $H(k[X_1]) = k[X_1]$.

Thus $H(K^R) \cong \Lambda_k(\Sigma H_1(K^R))$,

i.e. Koszul homology is generated by degree 1 elts.

- Then ② ⇒ ③ ⇒ ④ is clear.

  ↓ since $\Lambda_k(\Sigma H_1(K^R))$ has 0 differential.

④ ⇒ ① ☆ important criterion.

- By the above, $H_2(K^R) = (H_1(K^R))^2 \Leftrightarrow H_2(k[X]) = (H_1(k[X]))^2$

  Consider $H_1(k[X])$: we know $\partial(X_1) \subset m^2$ & $\partial(X_2) \subset mX_1$;

hence $H_1(k[X]) = kX_1$ since $Z_1 = X_1$, $B_1 = 0$.

- Then $H_2 = H_1^2 = kX_1^2$, but none of the variables $t \in X_2$ is a boundary since $t \notin (X_{\geq 1})^2$; yet kX_1^2 has no variable of deg 2 $\Rightarrow X_2 = \emptyset$.

  Thus $Q[X] = Q[X_1] \Rightarrow R$ is CI.

e.g. For $R = k[[x,y]]/(x^2, xy)$, $H_2(K^R) \neq (H_1(K^R))^2$ since:

$u \in H_2(K^R)$; $u \neq w_1 w_2 = 0$ and $w_1^2 = w_2^2 = 0 \Rightarrow u \notin (H_1(K^R))^2$.

4. Localization & Wiebe's Thm.

① Thm: R complete noetherian local. Then R complete intersection $\Rightarrow R_p$ complete intersection $\forall\, P \in \operatorname{Spec} R$.

pf:

- R complete $\Rightarrow R = Q/I$ for Q reg., I gen. by reg. seq.

✡ First we have $R_p = Q_{\mathfrak{P}}/IQ_{\mathfrak{P}}$, where $\mathfrak{P} = \pi^{-1}(P)$, $\pi: Q \to Q/I = R$.

- $Q_{\mathfrak{P}}$ is regular by chapter 0 (using pdim) ✡ ; $I = (\underline{f})$, then $f_1/1, \dots, f_n/1$ are also reg. seq. since $\begin{cases} Q_{\mathfrak{P}} \text{ flat} \Rightarrow \cdot f_i/1 \text{ is inj.} \\ I \subset \mathfrak{P} \text{ (since } \mathfrak{P} \in \operatorname{Spec}(Q/I)) \Rightarrow f_i\text{'s are not units.} \end{cases}$

- Then $R_p = Q_p/IQ_p$, but need $\hat{R}_p = \hat{Q}_p/I\hat{Q}_p$:
$\hat{Q}_p$ is regular & $I\hat{Q}_p$ reg. seq. since $\hat{Q}_p$ flat and $p\hat{Q}_p$ is the max. ideal.

This says localisation of complete rings doesn't change CI.

② Yet another homology criterion for CI.

Thm (Wiebe's): $n = \operatorname{embdim}(R) - \operatorname{depth}(R)$. (codepth)
Then R is CI $\Leftrightarrow H_1(K^R)^n \neq 0$.

Rmk: this is reasonable by depth sensitivity & $H(K^R)$ gen. by H_1.